

Survey Paper on IoT based Intrusion Detection System: Datasets and Techniques

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Abstract— As the pervasiveness of IoT devices in intelligent applications, traditional cybersecurity protection strategies cannot be directly applied. Additionally, the heterogeneity, dynamics, and scalability of these devices require sophisticated security and protection strategies. Motivated by these factors, we examine current security and privacy mechanisms to provide an overview of both academic and industrial domains and pave the way for further investigation. In this article, we discuss IoT devices, their uses, and how to protect these devices with an intrusion detection system developed using different datasets and different machine learning techniques. At the same time, it addresses existing intrusion detection systems which detect and classify various attacks in a timely and automated manner, especially with the continuously growth in number of attacks due to nature of data used in different IoT gadgets and its complexity of network.

Keywords— (*Internet of Things*) IoT, datasets, Intrusion detection System (IDS), smart devices, machine learning, attacks.

I. INTRODUCTION

The Internet of Things is a system that connects an ever-expanding coherent collection of specialized sensors and communication devices to more and more aspects of everyday life [1]. Basically, the idea of connecting an embedded device, computer, or sensor to the internet, either wirelessly or over a network. These gadgets can be supervised and controlled when they communicate with each other and share data over the Internet. Devices stated encompass the whole thing from cellular phones, domestic security, smart TVs, vehicles, and more. These are regularly known as linked or smart gadgets due to the fact they could speak thru a technique recognised as machine-to-machine (M2M) communication. These devices consist of three layers: recognition layer, network layer and application layer. At the sensory layer, sensors embedded in various fields capture data from their surroundings, but they have low processing power and low battery power, so they send data to devices on other layers over network media. At the network layer, data can be sent from the IoT sensor to other storage media devices via Bluetooth, BLE, RFID, etc. At the application layer, it can be processed and checked with a web interface developed in programming languages for instance Java, C#, Python etc.

These devices continue to grow in line with user needs, such as smart watches and smart meters. As such, each layer can be vulnerable to threats such as physical attacks, software intrusions, and network (IP address) hacking.

Various connected internet resources can be attacked anywhere in the world at any time, making cyber security an important issue. Cybersecurity revolves around three key security aspects as Confidentiality, Authentication and Data Protection [2]. Confidentiality is the safety of statistics in order that legal customers have get entry to private information. Authentication allows you to secure your network and ensure that best authenticated customer can get right of entry to blanketed resources. Access manipulate approach that best authorized customers can get right of entry to it.

There are number of security protocols that developed to ensure the proper protection of IoT Systems. These classified into reactive and proactive protocols. Proactive protocols developed to protect communication with authentication and encryption methods [3]. These pre-emptive logs can protect users from attacks that focus on sensitive statistics or statistics on another node, but compromised nodes can propagate modified statistics to surviving community nodes [4]. Reactive protocols have now been developed to detect and prevent from attacks called malicious activities.

Our contributions to this work are:

- To review attacks and mitigation methods for IoT devices in different fields.
- To review previous work on NIDS.
- The function of datasets and system mastering strategies in various intrusion detection systems.

This paper comprehensively describes the applications of IoT in various fields. In addition, classification of intrusion detection systems to protect IoT-based systems from diversity of attacks. The relaxation of the thing is prepared as follows: Section II provides a literature review. Section III describes special styles of intrusion detection systems. Section IV elaborate the datasets used, Section V reviewed on diverse machine gaining knowledge of strategies utilized in intrusion detection systems, and sooner or later Section VI provides the conclusions of the modern work.

II. RESEARCH METHODOLOGY

This section provides the methods used in this study. The methodology consists of planning and execution of the reviews. The methodology uses following steps.

- The formulation of key research questions.
- The formulation of search processes.

- Data extraction process and execution of analysis and classification.

A. In formulation of research question, it is further decomposed into many to achieve the objective of study. These are:

What is the use of primary study.

What are the methods utilised by the primary study.

What are the datasets used by the primary study.

What are the evaluation metrics used by the primary study.

B. In search process, data sources are used like IEEE Xplore, Science Direct, ACM, Springer and Wiley and queries are filtered using keywords IoT, attack classification, intrusion detection system, IoT based systems.

C. In Data Extraction, important data is collected for primary study and thus analysis is considered for each research question. Hence, classification of primary study is categorised with respect to each research question such as datasets, methods and evaluation metrics.

III. LITERATURE REVIEW

In this section, introduction of various techniques, methods and attack classification proposed for IoT. According to Yanchen Qiao et al.[5] suggest a way of malware class primarily based totally on Word2Vec which is malware characteristic illustration technique primarily based totally on herbal language processing and Multilayer Perception (MLP) and they follow deep mastering set of rules to classify the vectorized malware to assure the protection of IoT devices. They tested this method on BIG 2015 and classified malware in 9 families and check results that gives accuracy of 99.54%. Karim Lounis et al.[6] show the taxonomy of attacks and classified them into 3 varieties of assaults inclusive of fabrication assaults, domination assaults and interception assaults and number of attacks under these categories. Ruchi Vishwakarma et al.[7] discussed about the diversity of IoT attacks. Out of which they talked about Zero Day DDoS attack. Hence, used classification method to classify them and try to know the attackers IP address, MAC address, port number etc. from where they insert malicious code and authors try to know what type of information they want to know and what type of source code they transfer to machines. Then device is used to get primarily based totally detection framework to teach their classifiers effectively. Faisal Hussain, Syed Ghazanfar Abbas et al. [8] used CNN method which cannot efficiently work on text data or .csv file. So, they transform data sets (CICDDoS2019) into image using Fast Fourier Transformation (FFT) and formed into channel image matrix (C0 to C11) which include 11 types of DoS and DDoS attacks and after that they apply ResNet Model and this methodology gets 99.99% accuracy to detect DoS and DDoS attacks. R Gurunath et al.[9] elaborated different issues which can be occurred in IoT based Network. These issues can be power consumption, security issues as credentials based and malicious code which can stop working of IoT network. Espen Nystad et al.[10] show the pilot study in which they used the HAMMLAB (Halden man machine

laboratory) to detect the intrusion occurred when the pressure level exceeds values which was unknown to them but they found at last that there was an attack which alter its behaviour. Jing Wang et al.[11] discussed PDAM IoT based Smart grid system that may reap ideal facts confidentiality towards now no longer best a malicious aggregator however additionally a curious manage centre for a certified user support. Wissam Aoudi et al.[12] defined approximately the PADS as a way to cause an alarm each time the detection crosses predetermined threshold value. Mohammed Hayman et al.[13] proposed HFSDT-IoT which have two phases. In first phase, Ethereum PoS (Proof of Stake) Protocol used to discover DoS attacks. and after that in second phase cross-blockchain encryption used to execute inter blockchain transactions for safety.

Meenigi Ramesh Babu et al.[14] explained literature review of different methods used in previous year and postulated the Deep belief network (DBN)model that make accurate detection and classification and use hybridization of SLnO and WOA in Whale Sea Lion Algorithm (CWSA). Kehe Wu et al.[15] suggest a double hidden Markov model to identify anomalous behaviour of wireless power terminals. Using NS-2 Simulator, experimental results of anomalous behaviour identification and evaluation are analysed from which they calculate average attack time. Faik Kerem et al.[16] talked about different types of attacks after that they give tools through which attacks can be launched and discussed supervised machine learning models that deployed on edge node for Wi-Fi attack detection. Pankaj Thorat et al.[17] developed Intrusion Alarm System to identify malicious visitors from IoT-G. Noe Marcelo Yungaicela-Naula et al. [18] provide the implementation of a modular and bendy SDN-based structure to come across delivery and alertness layer DDoS assaults using more than one Machine Learning (ML) and Deep Learning (DL) models. Mohammad Dawood Momand et al.[19] proposed a unique Machine Learning-primarily based totally steady RPL routing (MLRP) protocol to effectively come across the assault behaviors. Table I shows the different attacks and mitigation methods for IoT devices and detail of their limitations is given.

TABLE I. LITERATURE ON IoT BASED ATTACKS AND METHODS WITH THEIR LIMITATIONS

Author	Year	Attack	Mitigation	Limitation
[5] Yanche n Qiao et al.	2021	Malware	Deep Learning Algorithm	Depth research in Malware Detection and classification is required.
[11] Jing Wang et al.	2021	Malicious data mining attack	PDAM, Elliptic Curve Cryptographic (ECC) mechanism	Needed high computation.
[12] Wissam Aoudi et al.	2021	Physical attack	PADS (Process Aware Detection System)	Time consuming process.
[13] Moham med Hayman et al.	2021	DoS such as sink-hole attack, grey- hole attack, black-hole attack	Ethereum PoS (Proof of Stake) Protocol, secp256k1 encryption with cross- blockchain	Less number of IoT devices used.
[14] Meenigi	2021	DoS and DDoS	Machine learning and	Comparison is not done between

Author	Year	Attack	Mitigation	Limitation	Author	Year	Attack	Mitigation	Limitation
Ramesh Babu et al.			Deep learning	methods.	[9] R Gurunath et al.	2018	DDoS attack	-	Only described issues which can occur in IoT network. Any technique is not mentioned to solve it.
[15] Kehe Wu et al.	2021	Abnormal Network Behavior	Double Hidden Markov Model (DHMM)	Does not compare results with other models					
[16] Faik Kerem et al.	2021	Mirari Botnet Attack, DoS attack, DDoS attack	Supervised machine Learning Models	Single network is used for attack detection.					
[17] Pankaj Thorat et al.	2021	Malicious traffic injection attack	Random Forest, CatBoost Machine Learning algorithm	Does now no longer upload exact effects approximately the effect of proposed alarm manager gadget on packet loss at IoT-G.					
[18] Noe Marcelo Yungaica-Naula et al.	2021	DoS, DDoS attack	kNN, SVM, RF, MLP, CNN, GRU, LSTM	Does not include Scalability component and mitigation module					
[19] Moham mad Dawood Moman d et al.	2021	DoS	MRLP classifier, SVM	The usage of deep studying fashions and ensemble classification approach has now no longer used to offer a correct protection answer for IoT scenarios.					
[6] Karim Lounis et al.	2020	Spoofing, Packet Replay, Packet Forging, Sniffing, Wi-Fi – wardriving, WEP packet Decryption, chop-chop Attack.	Symmetric Cryptography (3 DES) or Elliptic Curve Cryptography (ECC), AES Symmetric Cipher.	Does not cover long range and mid-range IoT wireless communication technologies.					
[10] Espen Nystad et al.	2020	Cyberattack	-	It's only a pilot study to find that there is an attack and not given any solution to mitigate the attack.					
[8] Faisal Hussain ,Syed Ghazanfar Abbas et al.	2020	DoS, DDoS	AI based techniques such as state-of-art Convolutional Neural Network (CNN) model called ResNet	Only worked for DDoS attack.					
[7] Ruchi Vishwakarma et al.	2019	IoT Malware also called as Zero -Day DDoS attack	Deep learning Models and Machine Learning Models based on Unsupervised Learning, Classification Algorithm	This approach is not broad.					

From Table I it is observed that most of the researches done to find attacks from IoT devices used in various fields using various deep learning methods. Researchers and academicians are using these techniques as these have strengths such as:

- Better self-learning capabilities.
- Works well with unstructured data.
- Scalability.
- Feature generation automation.
- Supports parallel and distributed algorithms.
- Advanced analytics.
- Cost effectiveness.

Due to these capabilities, above mentioned techniques get better accuracy and less processing time. It is discovered that there are exceptional intrusion detection structures that are used to locate the sort of assaults relying on their information units and machine gaining knowledge of algorithms which robotically classify the training of assaults like DoS, DDoS, Mirari, Botnet etc. Now in the next section, we will discuss intrusion detection systems.

IV. INTRUSION DETECTION IN IOT

Intrusion detection is the pastime of detecting movements achieved by intruders on facts systems. These movements are intrusion tries aimed at gaining unauthorized get entry to pc systems. Intruders may be outside or internal. Insiders are customers inside your community who've valid get admission to an experiment. External intruders are customers outdoor the goal community who are trying to find to benefit unauthorized get admission to system information, that allows you to benefit get admission to take advantage of unauthorized privileges [20]. IDS may be divided into community-primarily based totally IDS (NIDS) and host-primarily based totally IDS (HIDS). Network-primarily based totally IDS (NIDS) connects to 1 or extra created community segments to screen community site visitors for malicious activity. Host-primarily based totally IDS (HIDS) is hooked up to a laptop tool and monitors malicious hobby inside the gadget. Unlike NIDS, HIDS now no longer handiest analyses community traffic, however additionally jogging gadget calls, processes, report gadget changes, inter-method communication, and alertness logs [21]. Intrusion detection strategies fall into 4 categories, anomaly-based, signature-based, specification-based, and hybrid, relying at the detection mechanism used withinside the system. The purpose of this section is to explain how to develop an IDS for IoT using these techniques.

- A **Signature-based IDS:** In a signature-primarily based totally approach, an IDS is a device or community behavior that suits assault signatures saved in an inner IDS database. Alarms are brought about whilst gadget or community hobby suits a storage pattern. While

signature-primarily based totally IDS are correct and particularly powerful at detecting acknowledged threats, and their mechanisms are smooth to understand, this method has restrained fulfilment in detecting new or unknown variations of attacks. The correct signatures for these attacks are still unknown.

- B Anomaly-based IDS: Anomaly-based IDS compares a system's activity at a given point in time to its normal operating profile and generates alarms when deviations from normal behavior exceed thresholds. This technique is powerful in detecting new and unknown attacks, especially those related to resource abuse. However, all non-normal behaviors are considered interventions, and gaining knowledge of the overall variety of regular behaviors isn't smooth task [22].
- C Specification-based IDS: The intended behavior of network components is defined by a set of rules and standards called specifications. When a network disconnection occurs, it is detected by a specs-based approach. Behavior is out of specification. Therefore, specification-primarily based totally detection serves the equal cause as anomaly-primarily based totally detection. Discover anomalies by recognizing deviations from normative behavior is an exception. There is one vital distinction among those strategies. Specification-driven strategies require a human professional to manually outline the regulations for every method. Manually defined criteria typically have a lower false positive rate compared to anomaly-based detection. In addition, specification-based recognition systems can be operational immediately after specification is created, requiring no training. Manually defined specifications, on the other hand, are not adaptable to many situations.
- D. Hybrid IDS: To maximize the benefits while minimizing the drawbacks, the hybrid method uses anomaly-based detection, specification-based, and signature-based ideas. These intrusion detection systems process different data sets with different machine learning algorithms. In the subsequent sections, we introduce a few datasets and machine mastering techniques used to discover assaults in maximum IDSs.

V. INTRUSION DETECTION DATASETS

The assessment dataset performs vital role in validating the IDS method via way of means of permitting us to assess the capability of the proposed intrusive conduct detection method. The datasets used to assess community packs for business merchandise aren't with ease to be had because of protection concerns. However, there are numerous publicly to be had datasets, with ADFA-LD, NSL-KDD, DARPA, KDD and normally used as benchmarks. This section describes existing datasets used for IDS creation and comparative evaluation. Table II shows various datasets used in the literature review.

From Table II, attacks differ based on these data sets used by various intrusion detection systems. These records have several characteristics that are described below:

A DARPA / KDD Cup99: The first try to create an IDS dataset became made in 1998 via the Défense Advanced Research Project Agency. The collected community package was approximately 4 gigabytes in size and contained approximately 4,900,000 records. Looking at the two weeks of information, there were about 2 million connection records, each with 41 capacities, categorized as normal or abnormal. These records are outdated as they do not contain information about the latest malware attacks.

TABLE II. DATASETS USED IN LITERATURE REVIEW

Author	Year	Datasets	Techniques	Classification	Attack Classes
[29] Dominic Lightbody et al.	2022	Five different datasets naming Comp5, AIO, Comp 4, Comp 3, Comp 2, Comp 1	CNN	Multiclassification	Spike, Noise 1, Noise 2, Constant, Drift
[25] Tanzila Saba et al.	2022	NID Dataset and BoT-IoT datasets	Convolutional neural network (CNN)	Multiclassification	DoS, DDoS
[23] Imtiaz Ullah et al.	2021	IoT-IDS2020, and IoT-23, BoT-IoT, IoT Network Intrusion, MQTT-intrusion detection datasets.	CNN	Multiclass and binary classification	DoS, DDoS, Scan, Theft, Brute Force, MITM, Mirari
[28] Sudha Varalakshmi et al.	2021	MQTT-IoT-IDS2020, IoT Network Intrusion, IoT-DS, and BoT-IoT	CNN	Binary classification and Multiclassification	Botnets and DDoS attacks.
[24] Kaur et al.	2020	CICIDS2017 and CICIDS2018 datasets.	CNN	Multiclassification	Brute force, DoS attack, Web attack, Infiltration, SQL injection, DDoS attack, and Bot.
[26] Jiyeon Kim et al.	2020	CSE-CIC-IDS2018 and KDD CUP 1999	Convolutional Neural Network (CNN)	Binary classification and Multiclassification	DoS and DDoS attack.
[27] Mengmeng Ge et al.	2019	Bot-IoT dataset	Feed Forward Neural Network	Binary classification and Multiclassification	DoS and DDoS attack
[30] Yihan Xiao et al.	2019	KDD-CUP99 dataset	CNN model.	Multiclassification	DoS, Probe, U2R, R2L

For example, attacker behavior is unique to unique community topologies, working systems, software programs, and criminal toolkits. Nevertheless, KDD99 persisted for use as a benchmark inside his IDS studies community and remains utilized by researchers today.

B. CAIDA: This dataset consists of network traveller rows from a disbursed denial of service (DDoS) assault and turned into produced in 2007 (Hick et al., 2007). This kind of service denial assault tries to sabotage the target's pc or everyday traveller of the network via way of means of flooding it with a huge range of groups her packages. A drawback of the CAIDA data set is that it no longer contains a wide variety of attacks. Also, the records amassed now not consists of features from the network as a whole, making it hard to differentiate between abnormal and normal traveller flows.

C NSL-KDD: NSL-KDD is superior from the preceding Cup99 dataset KDD. Statistical evaluation of the Cup99 dataset revealed a widespread hassle that considerably impacted the accuracy of intruder detection and the consequences of deceptive checks of AIDS. A major problem in this data set is number of replica packages. The 2009 NSL-KDD dataset from the KDD Cup'99 dataset clarifies the above problems with the aid of using disposing of reproduction statistics. The NSL-KDD training data set contains 125,973 statistics and the test data set consists of 22,544 statistics. The NSL_KDD record contains 22 attacks for education and 41 attributes [31].

D CICIDS2017: The CICIDS2017 dataset contains information on all benign behaviors as well as the latest malware attacks. Includes, brute force SSH, DoS, heartbleed, brute force FTP, intrusions, botnets, DDoS, web attacks. Modems, firewalls, switches, routers, and extraordinary operating systems such as Apple's mac-OS, and open supply running systems Linux. This dataset consists of 80 community waved features collected from community traffic [32].

E CICIDS2018: This dataset is the result of a joint layout among the Dispatches Security Establishment (CSE) and the Canadian Institute for Cybersecurity (CIC). CIC makes use of the idea of biography to systematically derive cybersecurity records. It includes a detailed description of the intrusion along with an abstract distributed model of low-level operations, protocols, or network reality. This dataset contains seven different attack scripts. Videlicet Bruteforce, Heartbleed, botnets, DoS, DDoS, web attacks, external network intrusion. The attack structure includes 50 machines, the victim association has 420 of his PCs, his 5 departments including 30 waiters. This record contains the network business and loglines for each machine on the victim's side, as well as 80 network business functions that were wiped from hijacked companies using CICFlowMeter-V3 [33].

F. CICDDOS 2019: This is another new record, recording only DDoS attack cases on a topology of 87 functions. Attack scenarios documented in this dataset include UDP, LDAP, DNS, UDPlag, NetBIOS, portmap, SSDP, NTP, MSSQL, syn, and more. We sampled 129.88 for testing to get a reliable estimate of the recommended track. For more information on DDoS attack classification and testing, see [34]. The following segment describes diverse device

getting to know strategies utilized in intrusion detection systems.

VI. MACHINE LEARNING TECHNIQUES USED IN INTRUSION DETECTION SYSTEM

Machine learning is a subset of AI. It learns from expertise data and facilitates several utility domains such as computing, telecommunications and signal management processing. It solves complex algebraic problems and has proven high accuracy in finding attacks and give variety of cyber-security solutions. These algorithms are utilized in IDS to categorise conduct as ordinary or extraordinary via way of means of producing fashions able to sample popularity that predict intrusions. A challenge in IDS-based ML implementations is to create models with a low quantity of false positives and excessive detection accuracy. Given the diversified environment of IoT and its dynamics, Anomaly-based absolutely ML engineering can be a key solution to decorate detection of modern-day new advanced attacks and decorate classification performance. By extracting attributes from data-source, various ML methods can implement to classify data. The consequences may be utilized by IDS to make decisions. Classifier ML has two parts. These are training as well as testing part. Training part builds a model that can learn feature distributions and recognize patterns. During the testing part, we applied the model to detect anomalies [35-37]. Table III shows various techniques used in the literature.

TABLE III. TECHNIQUE USED IN LITERATURE

Author	Year	Review paper Implem entation	Dataset Used	Techniq ue Used	Technique results
[38] Sibi Amaran et al.	2021	IDS based on Optimal Support vector machine in WSN.	NSL KDDCu p 99	Optimal Support Vector Machine	Accuracy=94.31 TPR=95.67 FNR=4.33 Detection Rate=95.18
[39] Nelly Elsayed et al.	2021	BiLST M-CNN hybrid model	Dataset which available on the IEEE DataPort	BiLSTM -CNN	F1-Score= 98.90 Precision = 98.20 Accuracy=98.93 Recall =99.61
[40] Abhijit Dnyane shwar Jadhav et al.	2021	TP-IDS with two phases	NSL KDD	TP-IDS	Accuracy=99.93
[41] Monika D.Roka de et al.	2021	Identific ation of intruder s using machine learning algorithm	KDDCU P99 and NLSKD D	Naïve Bayes, SVM, Random Forest and ANN	SVM Precision=0.97 F-Score=0.95 Accuracy=0.95 Recall=0.94

Machine learning categorised into three parts: Unsupervised, semi-supervised and supervised. In current section, we create a classification of numerous ML strategies that may be used in the IDS context.

A. Supervised ML Approaches: A supervised approach is a predictive model advanced primarily based totally on a labelled training records set containing ordinary and odd records instances. The new data instance is compared to the model to determine which class it belongs to. Here number of supervised machine learning algorithms. These are K-nearest neighbors (KNN), linear classifiers, ANN and decision trees [42-43].

Logistic Regression: It is based on predictive analytics and used to perform analysis on binary type of dependent variable. This technique models binomial results. The result is either 1 or 0, acting a binary category of the high-quality elegance from the terrible elegance. We use a sigmoidal chord for output likelihood values and outperform classification [44].

Support Vector Machines: This method divides the gap into planes and unearths the hyperplane isolating them to categorise the data. New hidden statistics factors are then labelled primarily based totally on which aspect of the hyperplane they're on and only suitable for average-sized data sets having similar meaning features. The main advantage of this technology is its ability to perform actual-time intrusion detection and continuously alter training arrangements [45-46].

Naive Bayes: This is probability-based machine learning model that classifies using Bayes theorem. It learns parameters through thinking about that the cost of every characteristic is unbiased of the opposite functions given the magnificence variable. This class method is quicker in training as compared to the different linear classifiers and it is right for terribly huge datasets and large-dimensional data. But it offers bad results for linear classifier techniques [47-48].

K-Nearest Neighbor: This technique is used on small datasets and saves training dataset to build the model. It finds the nearest records factor withinside the education dataset to make predictions for brand new hidden records points [49-50].

Decision Tree: A selection tree is a flowchart-like shape wherein every internal node represents a check of an attribute. Each department represents a check result, every leaf node (that is, terminal node) represents a category label, and the route from the root (that is, the entire sample) to the leaf (that is, terminal node) represents a type rule. This method is straightforward to interpret. However, it's miles computationally extensive and is regularly especially erroneous and unstable (i.e., small adjustments withinside the statistics can result in huge adjustments withinside the shape of the most useful decision tree) [51].

Artificial Neural Network: The era is a brain-stimulated gadget that mimics the manner humans learn and neural networks are made from synthetic neurons referred to as perceptron. A neural community has an enter layer, an output layer, and a hidden layer that includes devices that remodel the enter right into an end result that may be used with the aid of using the output layer. This approach may be considered as a generalization of linear fashions that carry out more than one levels of processing to make decisions [52-53].

B. Unsupervised ML Approaches The unsupervised method does not compare specific labels with the training

part of dataset. Its objectives to find out about facts through modelling its shape and distribution. There are several unsupervised gadget studying algorithms, which include k-approach clustering, hidden markov models, and fuzzy logic [54].

K-means Clustering: The K-way clustering method has been utilized in WSN to detect intruders to beautify the safety of IoT systems [55-56]. This technique goals to generate okay clusters from a given information set with the aid of using repeatedly assigning every information factor to certainly considered one among okay clusters in keeping with the characteristics present. As a result, every cluster consists of samples with comparable characteristics and in effect, k-centroids are predicted that outline the facilities of the clusters. Each information factor is then assigned to the closest cluster centroid using the squared euclidean distance. The cluster centroid is then refigured by calculating via way of means of all samples associated with that cluster. These steps are repeated until no instances are available to modify the cluster. It is obvious that this process relies upon at the specification of the parameter k, which defines the cluster number, earlier than going for walks the algorithm [57].

Hidden Markov Model: These methods are probabilistic models with mathematical structure. These strategies are probabilistic fashions with mathematical structure. It is designed through a nation collection with markov residences and a remark collection wherein every image is emitted from the modern-day nation. HMMs can capture dependencies between successive sequences and are viewed as readable probability graph models. However, the computation is fairly complex and many of its parameters are freely estimated [58-59].

Fuzzy Logic: This method can handle uncertainty. As such, it is widely used to detect network threats. This is a handy manner to make choices primarily based totally on non-numerical and vague information. These depend on lexical inputs derived from membership functions to make decisions and membership function is an expression that used to define the fuzzy set to which a value belongs and the degree of membership in that set. The fuzzification operation of this technique maps mathematical input values to fuzzy club capabilities. However, the defuzzification operation maps the bushy output club capabilities to non-stop variables that could be used for selection purposes. This technique become utilized in reference to IDS. However, fuzzy good judgment isn't enough to come across all forms of attacks. It has to be blended with different classifiers for correct performance [60-61].

C. Semi-Supervised ML Approaches: In this approach, the training data item contain only regular class marks. The data instance has not been flagged as a bad class. Semi-supervised approaches used in machine learning because they can leverage available unlabelled data to enhance supervised learning tasks when labelled data are expensive or scarce [62],[64,65].

Hence, our finding from literature is given in Table IV.

We found that because the IoT networks are various day via way of means of day the form of assaults is likewise various as proven in above table. Hence, we can say that version in datasets is essential contribution of classifying one-of-a-kind classes of assaults.

TABLE IV. Findings in Literature

Author	Year	Dataset	Technique	Results	Attacks
[25] Tanzila Saba et al.	2022	NID Dataset and BoT-IoT datasets	Convolutional neural network (CNN)	Multiclassification	DoS, DDoS
[23] Imtiaz Ullah et al.	2021	IoT-23, IoT Network BoT-IoT Intrusion, MQTT-IoT-IDS2020	CNN	Multiclass and Binary classification	DoS, DDoS, Scan, Theft, Brute Force, MITM, Mirari
[24] Kaur et al.	2020	CICIDS2017 and CICIDS2018 datasets.	CNN	Multiclassification	DoS Attack, Brute force, SQL injection, Web attack, infiltration, DDoS attack, and Bot.

VII. CONCLUSION

IoT is a technology trend that enables new protocols such as applications and services. Many physical objects can be connected to the Internet, generating a large amount of network traffic. However, IoT traffic can be used to detect malicious threat activity. In fact, IoT systems have some security gaps and vulnerabilities. Most commonly, attackers abuse this new technology to compromise user privacy. In this article, a detailed literature review on various intrusion detection systems using different datasets and different machine learning methods such as supervised, unsupervised and semi-supervised methods. The existing literature is reviewed and various studies relevant to the research question are selected. It is found that there are intrusion detection models that have small datasets and the large number of IoT attack types and technologies used in intrusion detection systems can be divided into three subcategories based on the availability of tagged traffic data. Moreover, metrics such as storage efficiency, power consumption, and processing are not considered in most relevant studies when evaluating IoT systems. This work provides a framework for many other works in this research area. It is primarily used to find attacks in IoT-based systems. This section also describes attack classification methods used in the literature review. These datasets are old rather than new, so future work can use the latest datasets to determine the current status of the attack.

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